

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$u_3 = £20000 \times 1.08^2 = (£) 23328^*$	B1*	1.1b
		(1)	
(b)	$20000 \times 1.08^{n-1} > 65000$	M1	1.1b
	$1.08^{n-1} > \frac{13}{4} \Rightarrow n-1 > \frac{\ln(3.25)}{\ln(1.08)}$ or e.g. $1.08^{n-1} > \frac{13}{4} \Rightarrow n-1 > \log_{1.08}\left(\frac{13}{4}\right)$	M1	3.1b
	Year 17	A1	3.2a
		(3)	
(c)	$S_{20} = \frac{20000(1-1.08^{20})}{1-1.08}$ Awrt (£) 915 000	M1	3.4
		A1	1.1b
		(2)	
(6 marks)			
Notes			

(a)

B1*: Uses a correct method to show that the Profit in Year 3 will be £23 328. Condone missing units

E.g. $£20000 \times 1.08^2$ or $£20000 \times 108\% \times 108\%$

This may be obtained in two steps. E.g. $\frac{8}{100} \times 20000 = 1600$ followed by $\frac{8}{100} \times 21600 = 1728$ with the calculations $21600 + 1728 = 23328$ seen.

Condone calculations seen as $8\% \text{ of } 20000 = 1600$.

This is a show that question and the method must be seen.

It is not enough to state Year 1 = £21 600, Year 2 = £ 23 328



(b)

M1: Sets up an inequality or an equation that will allow the problem to be solved.

Allow for example N or n for $n-1$. So award for $20\,000 \times 1.08^{n-1} > 65\,000$, $20\,000 \times 1.08^n = 65\,000$ or $20\,000 \times (108\%)^n \geq 65\,000$ amongst others.

Condone slips on the 20 000 and 65 000 but the 1.08 o.e. must be correct

M1: Uses a correct strategy involving logs in an attempt to solve a type of equation or inequality of the form seen above. It cannot be awarded from a sum formula

The equation/inequality must contain an index of $n-1$, N , n etc.

Again condone slips on the 20 000 and 65 000 but additionally condone an error on the 1.08, which may appear as 1.8 for example

$$\text{E.g. } 20\,000 \times 1.08^n = 65\,000 \Rightarrow n \log 1.08 = \log \frac{65\,000}{20\,000} \Rightarrow n = \dots$$

$$\text{E.g. } 20\,000 \times 1.8^n = 65\,000 \Rightarrow \log 20\,000 + n \log 1.8 = \log 65\,000 \Rightarrow n = \dots$$

A1: Interprets their decimal value and gives the correct year number. Year 17

The demand of the question dictates that solutions relying entirely on calculator technology are not acceptable. BUT allow a solution that appreciates a correct term formula or the entire set of

calculations where you may see the numbers as part of a larger list

E.g. Uses, or implies the use of, an acceptable calculation and finds value(s)

for M1: $(n=16) \Rightarrow P = 20\,000 \times 1.08^{15} = \text{awrt } 63\,400$ or $(n=17) \Rightarrow P = 20\,000 \times 1.08^{16} = \text{awrt } 68\,500$ M1: $(n=16) \Rightarrow P = 20\,000 \times 1.08^{15} = \text{awrt } 63\,400$ and $(n=17) \Rightarrow P = 20\,000 \times 1.08^{16} = \text{awrt } 68\,500$

A1: 17 years following correct method and both M's

(c)

M1: Attempts to use the model with a correct sum formula to find the total profit for the 20 years.

You may see an attempt to find the sum of 20 terms via a list. This is acceptable provided there are 20 terms with $u_n = 1.08 \times u_{n-1}$ seen at least 4 times and the sum attempted.

Condone a slip on the 20 000 (e.g. appearing as 2 000) and/or a slip on the 1.08 with it being the same "r" as in (b). Do not condone 20 appearing as 19 for instance

A1: awrt £915 000 but condone missing unit

The demand of the question dictates that all stages of working should be seen. An answer without working scores M0 A0

Q2.

Question	Scheme	Marks	AOs
	$a = \left(\frac{3}{4}\right)^2 \text{ or } a = \frac{9}{16}$ or $r = -\frac{3}{4}$	B1	2.2a
	$\sum_{n=2}^{\infty} \left(\frac{3}{4}\right)^n \cos(180n)^\circ = \frac{\frac{9}{16}}{1 - \left(-\frac{3}{4}\right)} = \dots$	M1	3.1a
	$= \frac{9}{28}^*$	A1*	1.1b
		(3)	

Alternative 1:			
$\sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n \cos(180n)^{\circ} = \frac{-\frac{3}{4}}{1 - \left(-\frac{3}{4}\right)} = \dots \text{ or } r = -\frac{3}{4}$	B1	2.2a	
$\sum_{n=2}^{\infty} \left(\frac{3}{4}\right)^n \cos(180n)^{\circ} = -\frac{3}{7} - \left(-\frac{3}{4}\right)$	M1	3.1a	
$= \frac{9}{28}^*$	A1*	1.1b	
Alternative 2:			
$\sum_{n=2}^{\infty} \left(\frac{3}{4}\right)^n \cos(180n)^{\circ} = \left(\frac{3}{4}\right)^2 - \left(\frac{3}{4}\right)^3 + \left(\frac{3}{4}\right)^4 - \dots$	B1	2.2a	
$= \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^4 + \dots - \left(\frac{3}{4}\right)^3 - \left(\frac{3}{4}\right)^5 - \dots$ $\left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^4 + \dots = \left(\frac{3}{4}\right)^2 \left(\frac{1}{1 - \left(\frac{3}{4}\right)^2}\right) \text{ or } -\left(\frac{3}{4}\right)^3 - \left(\frac{3}{4}\right)^5 - \dots = -\left(\frac{3}{4}\right)^3 \left(\frac{1}{1 - \left(\frac{3}{4}\right)^2}\right)$ $\sum_{n=2}^{\infty} \left(\frac{3}{4}\right)^n \cos(180n)^{\circ} = \left(\frac{3}{4}\right)^2 \left(\frac{1}{1 - \left(\frac{3}{4}\right)^2}\right) - \left(\frac{3}{4}\right)^3 \left(\frac{1}{1 - \left(\frac{3}{4}\right)^2}\right)$	M1	3.1a	
$= \frac{9}{28}^*$	A1*	1.1b	
Alternative 3:			
$\sum_{n=2}^{\infty} \left(\frac{3}{4}\right)^n \cos(180n)^{\circ} = S = \left(\frac{3}{4}\right)^2 - \left(\frac{3}{4}\right)^3 + \left(\frac{3}{4}\right)^4 - \dots$	B1	2.2a	
$= \left(\frac{3}{4}\right)^2 \left(1 - \left(\frac{3}{4}\right) + \left(\frac{3}{4}\right)^2 - \dots\right) = \left(\frac{3}{4}\right)^2 \left(\frac{1}{4} + S\right) \Rightarrow \frac{7}{16}S = \frac{9}{64} \Rightarrow S = \dots$	M1	3.1a	
$= \frac{9}{28}^*$	A1*	1.1b	
(3 marks)			

Notes

B1: Deduces the correct value of the **first term** or the common ratio. The correct first term can be seen as part of them writing down the sequence but must be the **first term**.

M1: Recognises that the series is infinite geometric and applies the sum to infinity GP formula with $a = \frac{9}{16}$ and $r = \pm \frac{3}{4}$

A1*: Correct proof

Alternative 1:

B1: Deduces the correct value for the sum to infinity (starting at $n = 1$) or the common ratio

M1: Calculates the required value by subtracting the first term from their sum to infinity

A1*: Correct proof

Alternative 2:

B1: Deduces the correct value of the **first term** or the common ratio.

M1: Splits the series into “odds” and “evens”, attempts the sum of both parts and calculates the required value by adding both sums

A1*: Correct proof

Alternative 3:

B1: Deduces the correct value of the **first term**

M1: A complete method by taking out the first term, expresses the rhs in terms of the original sum and rearranges for “S”

A1*: Correct proof

Q3.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$24 + (6 \times 1.05) + (6 \times 1.05^2)$ minutes	M1	This mark is for a method to find the time taken for the competitor to run 6 km
	$= 96.915$ minutes $= 36$ minutes 55 seconds	A1	This mark is given for finding the total time as required
(b)	For example, 5th km $= 6 \times 1.05^1$ 6th km $= 6 \times 1.05^2$ 7th km $= 6 \times 1.05^3 \dots$ r th km $= 6 \times 1.05^{r-4}$	B1	This mark is given for showing the time taken to run the r th km, as required
(c)	$24 + \sum_{r=5}^{20} 6 \times 1.05^{r-4}$	M1	This mark is given for showing the total time to run the race is the time taken for the first 4 km added to the time taken from 5th to 20th km
	$= 24 + 6.3 \times \frac{(1.05^{16} - 1)}{1.05 - 1}$	M1	This mark is given for using $s = a \left(\frac{1 - r^n}{1 - r} \right)$ where $a = 6 \times 1.05 = 6.3$, $r = 1.05$ and $n = 20 - 4 = 16$
	$= 24 + 149.04$	A1	This mark is given for a correct total time (represented decimally)
	$= 173$ minutes and 3 seconds	A1	This mark is given for finding a correct total time given in minutes in seconds

Q4.

Question	Scheme	Marks	AOs
(a)	Uses $115 = 28 + 5d \Rightarrow d = (17.4)$	M1	3.1b
	Uses $28 + 2 \times "17.4" = \dots$	M1	3.4
	$= 62.8 \text{ (km h}^{-1}\text{)}$	A1	1.1b
		(3)	
(b)	Uses $115 = 28r^5 \Rightarrow r = (1.3265)$	M1	3.1b
	Uses $28 \times "1.3265^4" = \dots$ or $\frac{115}{"1.3265"}$	M1	3.4
	$= 86.7 \text{ (km h}^{-1}\text{)}$	A1	1.1b
		(3)	
(6 marks)			
Notes:			

(a)

M1: Translates the problem into maths using n^{th} term $= a + (n-1)d$ and attempts to find d

Look for either $115 = 28 + 5d \Rightarrow d = \dots$ or an attempt at $\frac{115-28}{5}$ condoning slips

It is implied by use of $d = 17.4$ Note that $115 = 28 + 6d \Rightarrow d = \dots$ is M0

M1: Uses the model to find the fastest speed the car can go in 3rd gear using $28 + 2"d$ or equivalent.

This can be awarded following an incorrect method of finding " d "

A1: 62.8 km/h Lack of units are condoned. Allow exact alternatives such as $\frac{314}{5}$

(b)

M1: Translates the problem into maths using n^{th} term $= ar^{n-1}$ and attempts to find r

It must use the 1st and 6th gear and not the 3rd gear found in part (a)

Look for either $115 = 28r^5 \Rightarrow r = \dots$ o.e. or $\sqrt[5]{\frac{115}{28}}$ condoning slips.

It is implied by stating or using $r = \text{awrt } 1.33$

M1: Uses the model to find the fastest speed the car can go in 5th gear using $28 \times "r^4"$ or $\frac{115}{"r"}$ o.e.

This can be awarded following an incorrect method of finding " r "

A common misread seems to be finding the fastest speed the car can go in 3rd gear as in (a).

Providing it is clear what has been done, e.g. $u_3 = 28 \times "r^2"$ it can be awarded this mark.

A1: awrt 86.7 km/h Lack of units are condoned. Expressions must be evaluated.

Q5.

Part	Working or answer an examiner might expect to see	Mark	Notes
(i)	$\sum_{r=4}^{\infty} 20 \times \left(\frac{1}{2}\right)^r$ $= \sum_{r=1}^{\infty} 20 \times \left(\frac{1}{2}\right)^r - (10 + 5 + 2.5)$	M1	This mark is given for a method to find the sum to infinity of a GP
	$= \frac{10}{1 - \frac{1}{2}} - (10 + 5 + 2.5)$	M1	This mark is given for a method to use a correct sum formula with a correct first term
	$= 2.5$	A1	This mark is given for a correct value for the sum
(ii)	$\sum_{n=1}^{48} \log_5 \left(\frac{n+2}{n+1}\right)$ $= \log_5 \frac{50}{49} + \log_5 \frac{49}{48} + \dots + \log_5 \frac{4}{3} + \log_5 \frac{3}{2}$	M1	This mark is given for writing out at least four terms of the sum, including the first two and the last two
	$= \log_5 \frac{3 \times 4 \times \dots \times 48 \times 49 \times 50}{2 \times 3 \times 4 \times \dots \times 48 \times 49} = \log_5 \frac{50}{2}$	M1	This mark is given for using the rules of logs and cancelling terms
	$= 2$	A1	This mark is given for a full proof to show the expression is equal to 2 as required
(Total 6 marks)			

Q6.

Question	Scheme	Marks	AOs
	(i) $\sum_{r=1}^{16} (3+5r+2^r) = 131\,798$; (ii) $u_1, u_2, u_3, \dots, u_{n+1} = \frac{1}{u_n}, u_1 = \frac{2}{3}$		
(i) Way 1	$\left\{ \sum_{r=1}^{16} (3+5r+2^r) = \right\} \sum_{r=1}^{16} (3+5r) + \sum_{r=1}^{16} (2^r)$	M1	3.1a
	$= \frac{16}{2}(2(8)+15(5)) + \frac{2(2^{16}-1)}{2-1}$	M1	1.1b
	$= 728 + 131\,070 = 131\,798^*$	A1*	2.1
	(4)		
(i) Way 2	$\left\{ \sum_{r=1}^{16} (3+5r+2^r) = \right\} \sum_{r=1}^{16} 3 + \sum_{r=1}^{16} (5r) + \sum_{r=1}^{16} (2^r)$	M1	3.1a
	$= (3 \times 16) + \frac{16}{2}(2(5)+15(5)) + \frac{2(2^{16}-1)}{2-1}$	M1	1.1b
	$= 48 + 680 + 131\,070 = 131\,798^*$	A1*	2.1
	(4)		
(i) Way 3	Sum = $10 + 17 + 26 + 39 + 60 + 97 + 166 + 299 + 560 + 1077 + 2106$ $+ 4159 + 8260 + 16457 + 32846 + 65619 = 131\,798^*$	M1	3.1a
		M1	1.1b
		M1	1.1b
		A1*	2.1
	(4)		
(ii)	$\left\{ u_1 = \frac{2}{3}, u_2 = \frac{3}{2}, u_3 = \frac{2}{3}, \dots \right\}$ (can be implied by later working)	M1	1.1b
	$\left\{ \sum_{r=1}^{100} u_r = \right\} 50\left(\frac{2}{3}\right) + 50\left(\frac{3}{2}\right) \text{ or } 50\left(\frac{2}{3} + \frac{3}{2}\right)$	M1	2.2a
	$= \frac{325}{3} \left(\text{or } 108\frac{1}{3} \text{ or } 108.3 \text{ or } \frac{1300}{12} \text{ or } \frac{650}{6} \right)$	A1	1.1b
	(3)		
(7 marks)			

Notes for Question	
(i)	
M1:	Uses a correct methodical strategy to enable the given sum, $\sum_{r=1}^{16} (3+5r+2^r)$ to be found Allow M1 for any of the following: - expressing the given sum as either $\sum_{r=1}^{16} (3+5r) + \sum_{r=1}^{16} (2^r)$, $\sum_{r=1}^{16} 3 + \sum_{r=1}^{16} (5r) + \sum_{r=1}^{16} (2^r)$ or $\sum_{r=1}^{16} 3 + 5\sum_{r=1}^{16} r + \sum_{r=1}^{16} (2^r)$ - attempting to find both $\sum_{r=1}^{16} (3+5r)$ and $\sum_{r=1}^{16} (2^r)$ separately (3×16) and attempting to find both $\sum_{r=1}^{16} (5r)$ and $\sum_{r=1}^{16} (2^r)$ separately
M1:	Way 1: Correct method for finding the sum of an AP with $a = 8$, $d = 5$, $n = 16$ Way 2: (3×16) and a correct method for finding the sum of an AP
M1:	Correct method for finding the sum of a GP with $a = 2$, $r = 2$, $n = 16$
A1*:	For all steps fully shown (with correct formulae used) leading to 131 798
Note:	Way 1: Give 2 nd M1 for writing $\sum_{r=1}^{16} (3+5r)$ as $\frac{16}{2}(8+83)$
Note:	Way 2: Give 2 nd M1 for writing $\sum_{r=1}^{16} 3 + \sum_{r=1}^{16} (5r)$ as $48 + \frac{16}{2}(5+80)$ or $48 + 680$
Note:	Give 3 rd M1 for writing $\sum_{r=1}^{16} (2^r)$ as $\frac{2(1-2^{16})}{1-2}$ or $2(2^{16}-1)$ or $(2^{17}-2)$

(i)	
Way 3	
M1:	At least 6 correct terms and 16 terms shown
M1:	At least 10 correct terms (may not be 16 terms)
M1:	At least 15 correct terms (may not be 16 terms)
A1*:	All 16 terms correct and an indication that the sum is 131 798
(ii)	
M1:	For some indication that the next two terms of this sequence are $\frac{3}{2}, \frac{2}{3}$
M1:	For deducing that the sum can be found by applying $50\left(\frac{2}{3}\right) + 50\left(\frac{3}{2}\right)$ or $50\left(\frac{2}{3} + \frac{3}{2}\right)$, o.e.
A1:	Obtains $\frac{325}{3}$ or $108\frac{1}{3}$ or $108.\dot{3}$ or an exact equivalent
Note:	Allow 1 st M1 for $u_2 = \frac{3}{2}$ (or equivalent) and $u_3 = \frac{2}{3}$ (or equivalent)
Note:	Allow 1 st M1 for the first 3 terms written as $\frac{2}{3}, \frac{3}{2}, \frac{2}{3}, \dots$
Note:	Allow 1 st M1 for the 2 nd and 3 rd terms written as $\frac{3}{2}, \frac{2}{3}, \dots$ in the correct order
Note:	Condone $\frac{2}{3}$ written as 0.66 or awrt 0.67 for the 1 st M1 mark
Note:	Give A0 for 108.3 or 108.333... without reference to $\frac{325}{3}$ or $108\frac{1}{3}$ or $108.\dot{3}$

Q7.

Question	Scheme	Marks	AOs
	Attempts $S_{\infty} = \frac{8}{7} \times S_6 \Rightarrow \frac{a}{1-r} = \frac{8}{7} \times \frac{a(1-r^6)}{1-r}$	M1	2.1
	$\Rightarrow 1 = \frac{8}{7} \times (1-r^6)$	M1	2.1
	$\Rightarrow r^6 = \frac{1}{8} \Rightarrow r = \dots$	M1	1.1b
	$\Rightarrow r = \pm \frac{1}{\sqrt{2}} \quad (\text{so } k = 2)$	A1	1.1b
(4 marks)			
Notes:			
M1: Substitutes the correct formulae for S_{∞} and S_6 into the given equation $S_{\infty} = \frac{8}{7} \times S_6$ M1: Proceeds to an equation just in r M1: Solves using a correct method A1: Proceeds to $r = \pm \frac{1}{\sqrt{2}}$ giving $k = 2$			